Table 51. Output voltage characteristics(1)

Symbol	Parameter	Conditions	Min	Max	Unit
V_{OL}	Output low level voltage	CMOS port(2) $\lvert I_{IO}\rvert$ = 8 mA $V_{DDIO1} \geq 2.7$ V	-	0.4	V
V_{OH}	Output high level voltage		V_{DD} - 0.4	-	V
V_{OL}(3)	Output low level voltage	TTL port(2) $\lvert I_{IO}\rvert$ = 8 mA $V_{DDIO1} \geq 2.7$ V	-	0.4	V
V_{OH}(3)	Output high level voltage		2.4	-	V
V_{OL}(3)	Output low level voltage	All I/Os $\lvert I_{IO}\rvert$ = 20 mA $V_{DDIO1} \geq 2.7$ V	-	1.3	V
V_{OH}(3)	Output high level voltage		V_{DD}- 1.3	-	V
V_{OL}(3)	Output low level voltage	$\lvert I_{IO}\rvert$ = 4 mA $V_{DDIO1} \geq 2.0$ V	-	0.45	V
V_{OH}(3)	Output high level voltage		V_{DD} - 0.45	-	V
V_{OLFM+}(3)	Output low level voltage for an FT I/O pin in FM+ mode	$\lvert I_{IO}\rvert$ = 20 mA $V_{DDIO1} \geq 2.7$ V	-	0.4	V
		$\lvert I_{IO}\rvert$ = 10 mA $V_{DDIO1} \geq 2.0$ V	-	0.4	

1. The I_{IO} current sourced or sunk by the device must always respect the absolute maximum rating specified in *Table 21: Voltage characteristics*. The sum of the currents sourced or sunk by all the I/Os (I/O ports and control pins) must always respect the absolute maximum ratings $\sum I_{IO}$.
2. TTL and CMOS outputs are compatible with JEDEC standards JESD36 and JESD52.
3. Specified by design. Not tested in production.

Input/output AC characteristics

The definition and values of input/output AC characteristics are given in *Figure 23* and *Table 52*, respectively.

Unless otherwise specified, the parameters given are derived from tests performed under the ambient temperature and supply voltage conditions summarized in *Table 24: General operating conditions*.

Table 52. I/O AC characteristics

Speed	Symbol	Parameter(1)(2)	Conditions	Min	Max	Unit
00	Fmax	Maximum frequency	C=50 pF, 2.7 V ≤ V_{DDIO1} ≤ 3.6 V	-	2	MHz
			C=50 pF, 2.0 V ≤ V_{DDIO1} ≤ 2.7 V	-	0.35	
			C=10 pF, 2.7 V ≤ V_{DDIO1} ≤ 3.6 V	-	3.00	
			C=10 pF, 2.0 V ≤ V_{DDIO1} ≤ 2.7 V	-	0.45	
	Tr/Tf	Output rise and fall time(3)	C=50 pF,2.7 V ≤ V_{DDIO1} ≤ 3.6 V	-	100.00	ns
			C=50 pF, 2.0 V ≤ V_{DDIO1} ≤ 2.7 V	-	225.00	
			C=10 pF, 2.7 V ≤ V_{DDIO1} ≤ 3.6 V	-	75.00	
			C=10 pF, 2.0 V ≤ V_{DDIO1} ≤ 2.7 V	-	150.00	